Keelan O'Doherty
17 February, 2026

Mechanical Engineering Intern
Burns McDonnell
Phoenix, AZ

Dear Hiring Manager,

My interest in engineering systems began long before college, but it is through my undergraduate research and design experience that this curiosity has grown into a clear commitment to applying engineering principles to real-world infrastructure challenges. I am currently a senior pursuing a B.S. in Aerospace Engineering at The University of Alabama Honors College, and I will be enrolling in a graduate program in Aerospace or Mechanical Engineering next year. I am enthusiastic about contributing to your team's work in planning, designing, and optimizing complex engineering systems that support critical infrastructure.

My work in aerodynamics and flight mechanics has shaped the way I understand complex systems, specifically how theoretical principles translate into functional designs and measurable performance. I applied this mindset in the Alabama Aerodynamics Research Laboratory (AAERO), where I helped design and experimentally test toroidal and planar UAV propellers. Working across CAD modeling, CFD simulations, prototype fabrication, and Python-based analysis taught me how to evaluate a system from concept through validation. That structured, analytical workflow directly aligns with the detailed design, evaluation, and updating of engineering systems described in this role.

My experience with the Vertical Flight Society further strengthened my systems-level approach. While developing a fixed-wing VTOL aircraft, I used MATLAB modeling and refined actuation logic to correct instability during the transition between flight modes. Solving this challenge required careful analysis, iterative troubleshooting, and collaboration across disciplines—skills that are essential when supporting senior engineers on large-scale planning and design projects.

Together with my broader engineering experience, including mechanical restoration projects, entrepreneurial project management, and hands-on fabrication work, these roles have prepared me to contribute meaningfully to a team environment focused on reliability, precision, and performance. The opportunity to apply my analytical and design background to power systems, transmission infrastructure, or consulting-based engineering projects is exactly the type of environment in which I am eager to grow and contribute.

Sincerely,
Keelan O'Doherty